Continuous Random Variables

ECON 97: Economic Toolkit

Patrick Molligo

UCLA

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iClicker course link

https://join.iclicker.com/RDNG

Quiz

Consider random variables X_1 and X_2 with joint PMF:

$$f(x_1, x_2) = \frac{x_2 - x_1}{6}$$

where $x_1 = -1, 1$ and $x_2 = 1, 2$. Find $E[X_2]$.

A. 4/3

B. 5/3

C. 2/3

D. 7/3

Joint Distributions: Recap

Joint PMF: $f(x_1, x_2) = P(X_1 = x_1, X_2 = x_2)$

Marginal PMF: $f(x_1) = P(X_1 = x_1)$ amd $f(x_2) = P(X_2 = x_2)$

$\longrightarrow$ if X_1 & X_2 independent, then $f(x_1, x_2) = f(x_1)f(x_2)$

Covariance: $E(X_1X_2) - E(X_1)E(X_2)$

where $E(X_1X_2) = \sum_{x_1}\sum_{x_2}(x_1x_2)f(x_1, x_2)$

Correlation: $\rho_{X,Y} = \frac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}} \in [-1, 1]$

$\longrightarrow$ if X_1 & X_2 independent, then covariance and correlation are 0

Continuous random variables

The **support** of a variable determines distribution type:

If X is "countable", then it's **discrete**:

$$\text{E.g.,} \;\; X = 0, 1, 2, 3$$

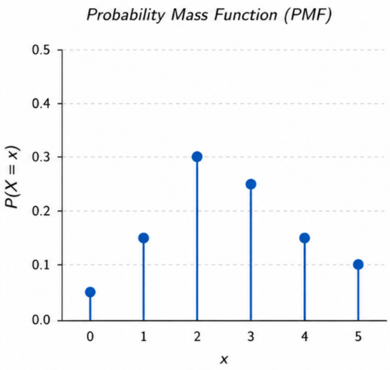
If X belongs to a range of the real number line, then it's **continuous**

$$\text{E.g.,} \;\; 0 \leq X \leq 3$$

For the example above, the continuous X can take the value 2.5, while the discrete can be 2 or 3 , but never 2.5

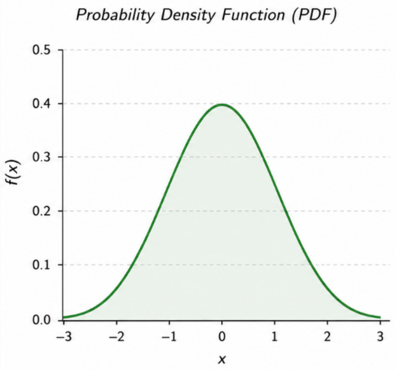
Discrete (PMF) vs. Continuous (PDF)

Discrete Variable



Example: *Number of heads in 3 coin flips*

Continuous Variable



Example: *Height of an adult (in cm)*

*Note: We call it a PDF for continuous ("density" vs. "mass")

Properties of continuous variables

Most of the usual rules apply, just replace $\sum$ with $\int$:

- $f(x)$: PDF of X, $f(x) \geq 0$
- $F(x) = P(X \leq x)$: CDF of X
- $\int_{-\infty}^{\infty} f(x)dx = 1$
- $E[X] = \mu_X = \int_{-\infty}^{\infty} xf(x)dx$
- $\text{Var}(X) = \sigma_X^2 = \int_{-\infty}^{\infty} x^2 f(x)dx - \mu_X^2$
  - Note: if support of X is $a \leq X \leq b$, then $\int_a^b$ gives the same results

Computing probabilities

New rules that apply only to continuous r.v.'s:

The derivative of the CDF is the PDF:

$$F'(x) = f(x)$$
$$F(x) = \int_{-\infty}^{x} f(t)dt$$

Probabilities are always over ranges:

$$P(a \leq X \leq b) = \int_{b}^{a} f(x)dx = F(b) - F(a)$$

$$P(X = a) = 0 \text{ (beware of trick questions!)}$$

Strict inequalities don't matter:

$$P(X \leq a) = P(X < a)$$

Continuous R.V. practice

$$f(x) = x^3/4, \quad 0 < x \leq 2$$

$$A_1 = \{x : 0 < x < 1\}$$

$$A_2 = \{x : 1/2 < x < 3/2\}$$

Compute:

- $P(A_1)$
- $P(A_2)$
- $P(A_1 \cup A_2)$

Continuous R.V. practice

Continuous R.V. practice

$$f(x) = c(x^3 + 8)/32, \quad -2 < x < 2$$

- Find c
- Compute mean and variance of X

Continuous R.V. practice

Special distribution: Uniform

Uniform: every outcome has equal probability

$$X \sim Unif(a, b), \quad a \leq x \leq b$$

$$f(x) = \frac{1}{b-a}$$

$$F(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leq x \leq b \\ 1, & x > b \end{cases}$$

$$E[X] = \frac{a+b}{2} \quad Var(X) = \frac{(b-a)^2}{12}$$

Special distribution: Normal

Normal distribution: classic bell curve, symmetric around mean

$$X \sim N(\mu, \sigma^2), \quad -\infty < x < \infty$$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Mean (μ) and variance (σ^2) will always be given to you

Special case: $\mu = 0$ and $\sigma^2 = 1$; called "standard normal"

$$Z \sim N(0, 1)$$

Use symbol $\Phi(\cdot)$ for CDF of Z

$F(x) = \Phi(x) = P(Z \leq x)$